

Unit Propagation

DSCI 220 · Lecture 7 activity

What to do

Every group has a **different** argument. There is no one right answer across the room — that is the point.

You are asked whether a conclusion **follows** from some premises. Rather than chase it forwards, we assume it does **not**: the negated conclusion joins the premises, and we look for a contradiction. If the premises plus the *denial* of the conclusion cannot all hold, then the conclusion followed all along.

Everything has already been turned into clauses for you.

The loop. Repeat until you cannot:

1. Find a clause with **exactly one literal**. That literal must be true. **Circle it** — call it a **unit**.
2. Look for that same literal in any other clause. That clause is already satisfied, so **circle the literal there too, and cross the whole clause out**. The circle records why it went.
3. In every clause containing the **negation** of your unit, strike out just that one literal. The rest of the clause stays.
4. If a clause loses its **last** literal, **circle what is left of it**. That is the **empty clause** — a clause with no way left to be true, so it is **False**. You have found a contradiction, and that is the best thing that can happen to you.

A circle always marks something you have **established**: a literal that must be true, or — best of all — an empty clause. A crossed-out clause always has a circle in it saying which literal did the work.

Three ways to stop, and each says something different about the argument:

You stopped because	So the conclusion
you found a contradiction no clauses were left	follows . Denying it was impossible does not follow — and the units you circled are a counterexample
no clause had a single literal	<i>cannot be settled this way</i> . Propagation alone is not enough

Set 1

The argument

1	D
2	$D \rightarrow \neg C$
3	$\neg C \wedge \neg B \rightarrow \neg E$
4	$\neg B$
5	$\neg G \vee \neg F$
$\therefore$	$\neg E$

Does that conclusion follow? To find out, we assume it does **not** — so E joins the premises — and look for a contradiction.

	Clause	Working
1	$\{D\}$	
2	$\{\neg C, \neg D\}$	
3	$\{B, C, \neg E\}$	
4	$\{\neg B\}$	
5	$\{\neg F, \neg G\}$	
6	$\{E\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg E$ follow from the premises?

How do you know you are finished?

Set 2

The argument

1	$\neg B$
2	$\neg B \rightarrow G$
3	$G \wedge \neg C \rightarrow \neg F$
4	$\neg C$
5	$D \vee \neg A$
$\therefore$	$\neg F$

Does that conclusion follow? To find out, we assume it does **not** — so F joins the premises — and look for a contradiction.

	Clause	Working
1	$\{\neg B\}$	
2	$\{B, G\}$	
3	$\{C, \neg F, \neg G\}$	
4	$\{\neg C\}$	
5	$\{\neg A, D\}$	
6	$\{F\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg F$ follow from the premises?

How do you know you are finished?

Set 3

The argument

1	B
2	$B \rightarrow E$
3	$E \wedge \neg A \rightarrow \neg F$
4	$\neg A$
5	$D \vee \neg C$
$\therefore$	$\neg F$

Does that conclusion follow? To find out, we assume it does **not** — so F joins the premises — and look for a contradiction.

	Clause	Working
1	$\{B\}$	
2	$\{\neg B, E\}$	
3	$\{A, \neg E, \neg F\}$	
4	$\{\neg A\}$	
5	$\{\neg C, D\}$	
6	$\{F\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg F$ follow from the premises?

How do you know you are finished?

Set 4

The argument

1	C
2	$C \rightarrow \neg D$
3	$\neg D \wedge \neg B \rightarrow A$
4	$\neg B$
5	$G \vee F$
$\therefore$	A

Does that conclusion follow? To find out, we assume it does **not** — so $\neg A$ joins the premises — and look for a contradiction.

	Clause	Working
1	$\{C\}$	
2	$\{\neg C, \neg D\}$	
3	$\{A, B, D\}$	
4	$\{\neg B\}$	
5	$\{F, G\}$	
6	$\{\neg A\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does A follow from the premises?

How do you know you are finished?

Set 5

The argument

1	$\neg D$
2	$\neg D \rightarrow G$
3	$G \wedge E \rightarrow C$
4	E
5	$F \vee \neg A$
$\therefore$	C

Does that conclusion follow? To find out, we assume it does **not** — so $\neg C$ joins the premises — and look for a contradiction.

	Clause	Working
1	$\{\neg D\}$	
2	$\{D, G\}$	
3	$\{C, \neg E, \neg G\}$	
4	$\{E\}$	
5	$\{\neg A, F\}$	
6	$\{\neg C\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does C follow from the premises?

How do you know you are finished?

Set 6

The argument

1	$\neg G$
2	$\neg G \rightarrow A$
3	$A \rightarrow \neg B$
4	$D \rightarrow C$
$\therefore$	D

Does that conclusion follow? To find out, we assume it does **not** — so $\neg D$ joins the premises — and look for a contradiction.

	Clause	Working
1	$\{\neg G\}$	
2	$\{A, G\}$	
3	$\{\neg A, \neg B\}$	
4	$\{C, \neg D\}$	
5	$\{\neg D\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does D follow from the premises?

How do you know you are finished?

Set 7

The argument

1	G
2	$G \rightarrow \neg D$
3	$\neg D \rightarrow A$
4	$\neg C \rightarrow F$
$\therefore$	$\neg C$

Does that conclusion follow? To find out, we assume it does **not** — so C joins the premises — and look for a contradiction.

	Clause	Working
1	$\{G\}$	
2	$\{\neg D, \neg G\}$	
3	$\{A, D\}$	
4	$\{C, F\}$	
5	$\{C\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg C$ follow from the premises?

How do you know you are finished?

Set 8

The argument

1	F
2	$F \rightarrow \neg A$
3	$\neg A \rightarrow B$
4	$E \rightarrow \neg G$
$\therefore$	E

Does that conclusion follow? To find out, we assume it does **not** — so $\neg E$ joins the premises — and look for a contradiction.

	Clause	Working
1	$\{F\}$	
2	$\{\neg A, \neg F\}$	
3	$\{A, B\}$	
4	$\{\neg E, \neg G\}$	
5	$\{\neg E\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does E follow from the premises?

How do you know you are finished?

Set 9

The argument

1	B
2	$B \rightarrow D$
3	$D \rightarrow A$
4	$F \rightarrow C$
$\therefore$	F

Does that conclusion follow? To find out, we assume it does **not** — so $\neg F$ joins the premises — and look for a contradiction.

	Clause	Working
1	$\{B\}$	
2	$\{\neg B, D\}$	
3	$\{A, \neg D\}$	
4	$\{C, \neg F\}$	
5	$\{\neg F\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does F follow from the premises?

How do you know you are finished?

Set 10

The argument

1	$\neg A$
2	$\neg A \rightarrow C$
3	$\neg B \vee \neg E$
4	$F \vee \neg G$
5	$\neg B \wedge F \rightarrow \neg D$
$\therefore$	$\neg D$

Does that conclusion follow? To find out, we assume it does **not** — so D joins the premises — and look for a contradiction.

	Clause	Working
1	$\{\neg A\}$	
2	$\{A, C\}$	
3	$\{\neg B, \neg E\}$	
4	$\{F, \neg G\}$	
5	$\{B, \neg D, \neg F\}$	
6	$\{D\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg D$ follow from the premises?

How do you know you are finished?

Set 11

The argument

1	$\neg E$
2	$\neg E \rightarrow F$
3	$B \vee \neg C$
4	$\neg D \vee G$
5	$B \wedge \neg D \rightarrow \neg A$
$\therefore$	$\neg A$

Does that conclusion follow? To find out, we assume it does **not** — so A joins the premises — and look for a contradiction.

	Clause	Working
1	$\{\neg E\}$	
2	$\{E, F\}$	
3	$\{B, \neg C\}$	
4	$\{\neg D, G\}$	
5	$\{\neg A, \neg B, D\}$	
6	$\{A\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg A$ follow from the premises?

How do you know you are finished?

Set 12

The argument

1	$\neg G$
2	$\neg G \rightarrow E$
3	$\neg B \vee \neg D$
4	$C \vee \neg A$
5	$\neg B \wedge C \rightarrow \neg F$
$\therefore$	$\neg F$

Does that conclusion follow? To find out, we assume it does **not** — so F joins the premises — and look for a contradiction.

	Clause	Working
1	$\{\neg G\}$	
2	$\{E, G\}$	
3	$\{\neg B, \neg D\}$	
4	$\{\neg A, C\}$	
5	$\{B, \neg C, \neg F\}$	
6	$\{F\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg F$ follow from the premises?

How do you know you are finished?

Set 13

The argument

1	C
2	$C \rightarrow A$
3	$\neg G \vee F$
4	$B \vee D$
5	$\neg G \wedge B \rightarrow \neg E$
$\therefore$	$\neg E$

Does that conclusion follow? To find out, we assume it does **not** — so E joins the premises — and look for a contradiction.

	Clause	Working
1	$\{C\}$	
2	$\{A, \neg C\}$	
3	$\{F, \neg G\}$	
4	$\{B, D\}$	
5	$\{\neg B, \neg E, G\}$	
6	$\{E\}$ (<i>negated conclusion</i>)	

Units you forced, in order:

How did you stop? found a contradiction ran out of clauses stalled

So does $\neg E$ follow from the premises?

How do you know you are finished?
